

Exercise 2: SQL FoundationsSQL Aggregate Functions & Operators(01) The students Table

(1) `SELECT DISTINCT department`
`FROM students`

department
IT
HR
Finance

(2) `SELECT department,`
`AVG(age) AS avg_age`
`FROM students`
`GROUP BY department;`

department	avg_age
IT	20.5
HR	22.0
Finance	23.0

(3) `SELECT department,`
`COUNT(*) AS student_count`
`FROM students`
`GROUP BY department`
`HAVING COUNT(*) > 1;`

department	student - count
IT	2
HR	2

```
(4) SELECT student_id,
        name,
        age,
        department
FROM students
WHERE age BETWEEN 21 AND 23
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

```
(5) SELECT student_id
        name,
        age,
        department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

(02) The courses Table

```
(06) SELECT department,  
       sum(credits) AS total_credits  
FROM   courses  
GROUP BY department  
HAVING sum(credits) > 5;
```

department	total_credits
IT	11

```
(7) SELECT course_id,  
          course_name,  
          department,  
          credits  
FROM   courses  
WHERE  credits != 4
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

```
(8) SELECT course_id,  
          course_name,  
          credits  
FROM   courses  
ORDER BY credits DESC  
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basic	3

(03) The enrollments Table

(9) `SELECT MAX(grade) AS max_grade,
MIN(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM enrollments;`

max_grade	min_grade	avg_grade
90	78	84.6

(10) `SELECT course_id,
COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;`

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(04) The salaries Table

(11) SELECT department,
sum(salary) AS total_salary,
sum(bonus) AS total_bonus,
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
FINANCE	70 000	6 000

(12) SELECT department,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000

department	avg_salary
IT	61 000
FINANCE	70 000

(13) SELECT employee_id,
name,
salary,
bonus,
(salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60 000

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

(15) The projects Table

```

(14) SELECT department,
        sum(budget) AS total_budget,
        avg(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING avg(budget) > 70000;

```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

```

(15) SELECT project_id,
        project_name,
        department,
        budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department != 'Marketing';

```

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000